

AI Usage Documentation

In compliance with ISF AI Guidelines (Type 2)

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1 Introduction

This document formally discloses the utilization of generative Artificial Intelligence (AI) during the development of the *HumanMusic* project. In accordance with the ISF AI Guidelines, the support provided for resolving the iOS and iPadOS Safari Web Audio and Canvas rendering issues is classified as **Type 2 – Generative Writing (Fixing/modifying source code with generative AI)**.

2 AI Tool Documentation (Type 2)

The following table details the AI tool used, the affected files, and a reflection on the tool's effectiveness:

Category	Details
Tool Name & Version	Google Gemini (July 2026)
AI Usage Classification	Type 2 – Generative Writing
Where it was used	<code>index.html</code> , <code>visualizer.js</code> , <code>player.js</code> , <code>server.js</code>
Sentiment / Reflection	The tool successfully identified the root causes of the iOS-specific Web Audio and Canvas bugs. It took a few iterations to navigate Apple's incredibly strict autoplay and source-swapping policies. Ultimately, it provided a highly robust solution using the Fetch API and Web Audio API directly, completely resolving the cross-device compatibility issues without requiring intrusive UI changes. I opted to use generative AI for this specific task because time was short and the bugs were exclusively isolated to Apple devices, which required complex, platform-specific workarounds.

Table 1: Overview of Google Gemini Usage

2.1 Chat History and Prompt Summary

Since a raw transcript of the conversation contains multiple file uploads, the key prompts and AI outputs are summarized below:

- 1. Initial Prompt (iOS Blockage):** *"I use the Web Audio API, Canvas API, and Audio/Video files on my website. The problem is that it doesn't work on iOS devices. There*

are solutions on the internet, but I don't want to write that much code to create a button and click it when needed... Help me solve the problem as simply as possible."

2. **Follow-up Prompt 1 (Canvas Rendering Bug):** [User provided a video demonstration showing that the canvas visualizer failed to render on iPhone.]
3. **Follow-up Prompt 2 (iPad-specific Freeze):** *"Why doesn't the visualizer show up on the iPad? The video works fine, but the visualizer doesn't."*
4. **Follow-up Prompt 3 (Audio Channel Block):** *"The problem now is that there is no sound on iOS devices and the visualizer is a straight line (stagnant)."*
5. **Follow-up Prompt 4 (Source Swapping Crash):** *"It only works on all devices except iOS. Can you explain what you found out and how we can solve this? Why won't it work on iOS anymore?"*

6. **AI Outputs and Solutions Provided:**

- **Global Audio Unlocker:** Implemented an invisible `unlockAudio` function that triggers on the first user touch (`touchstart` / `click`) to unlock the Web Audio `AudioContext`.
- **Bypassing HTML5 Audio Restrictions:** Identified that iOS Safari blocks source swapping (`audio.src = newUrl`) on previously unlocked media elements.
- **Web Audio API Direct Playback:** Re-architected `visualizer.js` to bypass the `<audio>` tag completely. Implemented `fetch()` and `AudioContext.decodeAudioData()` to load and play audio natively via `AudioBufferSourceNode`, ensuring 100% compatibility with Apple's policies.
- **Attribute Cleanup:** Removed `autoplay` and `crossorigin` attributes from `index.html` to prevent CORS errors on desktop browsers and immediate blockage on mobile Safari.
- **Robust Canvas Rendering:** Wrapped the `drawWaveform()` logic in conditional checks (`if (analyser)`) and provided fallback values in `resizeCanvas()` to prevent fatal script halts if audio data is temporarily unavailable.